ERIK LAITALA

R&D ENGINEER — MECHATRONICS & AUTONOMOUS SYSTEMS

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PROFESSIONAL SUMMARY

Electrical and mechatronics engineer with hands-on experience in electrical design, embedded integration, HIL test-system development, and autonomous platforms, spanning requirements, component selection, prototyping, validation, and real-world iteration. Named inventor on a granted patent for MPC-based autonomous marine docking.

TECHNICAL SKILLS

Programming & Control: Python, C++, Java, ROS2, MATLAB, Simulink, model predictive control, autonomous systems

Electrical & Embedded: KiCad, Embedded systems, sensors, actuators, HIL simulation systems, electrical design, system testing, technical documentation

Mechanical & Prototyping: SolidWorks, 3D printing, welding, fabrication, assembly, component selection

Design & Office Tools: LaTeX, Adobe InDesign, Adobe Illustrator, Excel

PROFESSIONAL EXPERIENCE

Electrical Design Engineer — Akkodis Electrical Solutions

January 2026–Present

- Developed and built hardware-in-the-loop (HIL) simulation systems as part of the HIL team at Scania, supporting testing and validation of embedded electronic systems
- Led a cross-functional RFP and proposal project for an international customer, coordinating electrical engineering, mechanics, firmware/software, production, suppliers, and stakeholders
- Structured requirements, technical concepts, prototype planning, component selection, BOMs, and technical documentation
- Developed an internal software tool that reduced a production-testing workflow from 60 to 6 minutes by automating measurement capture and Excel data logging, while integrating foot-pedal controls for efficient operator input

Research Engineer — Swedish Defence Research Agency (FOI)

November 2024–December 2025

- Developed and tested autonomous systems and advanced control methods for unmanned platforms
- Designed, 3D-printed, and built unmanned vehicles integrating hardware and software as test platforms for newly developed algorithms
- Led the autonomy lab during the lab manager's parental leave, including onboarding, demonstrations, equipment management, and component procurement
- Presented technical developments from the autonomy lab to approximately 400 colleagues

EDUCATION

MSc, Systems Control and Mechatronics — Chalmers University of Technology

2022–2024

- Relevant courses: robust and nonlinear control, deep machine learning, model-based development of cyber-physical systems, future renewable-based power systems, electrical machines
- Thesis: Autonomous Docking with Model Predictive Control; patent granted May 2026. Patent number SE2450627-1

BSc, Electrical Engineering — Chalmers University of Technology

2018–2021

- Relevant courses: object-oriented programming, circuit analysis, engineering measurements, electric power engineering
- Thesis: Secure and Stable Energy — Design and Construction of a Sustainable, Portable, Self-Starting Power Generator

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

Mechanical Technician and Graphic Designer — Chalmers Cortège Committee (annual school parade)

January 2021–December 2022

- Responsible for the construction of 20 motorized vehicles, including engine set-up, repair, welding, and assembly
- Produced a 100-page sponsorship publication, leading ideation, material sourcing, sponsor coordination, layout, and image creation using Adobe tools, taking the project from concept to print in 30 days

Secretary and PR — Board of the Student Union, Electrical Engineering Program, Chalmers

January–December 2020

- Represented 300 students in negotiations on courses, committees, student health, and facilities
- Facilitated communication between students and staff; helped integrate Chalmers' new Medical Engineering bachelor's program

LANGUAGES & ADDITIONAL

Swedish — Native · English — Fluent/professional · Driving licence, class B